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A Wide Comparison of Public Lossless Compressors on BF16 Model Weights

Evidence-grounded FCN academic paper

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Abstract

We report a wide comparison of public lossless compressors applied to BF16 weights of a publicly released large language model within the Forlais Compression Notation (FCN) research programme. The comparison covers more than sixty configurations spanning the DEFLATE, LZMA, ZSTD, brotli, and ZPAQ-style families, with and without standard public preconditioners such as bit-transpose, byte-swap, bit-plane separation, and exponent grouping. The best configuration in the comparison

attains a reported compression ratio of approximately $1.5085\times$ on the tested setting, which is approximately 99.1% of the FCN single-model entropy ceiling of approximately $1.522\times$ reported in a companion entry. No tested configuration exceeds the entropy ceiling. We interpret the comparison as direct evidence that the binding constraint in the single-model regime is the per-value entropy of the trained-weight distribution and not the choice of public compressor. The claim is scoped to the tested setting and does not extend to cross-model conditioning, learned non-IID coders, format redesign, or task-specific model compression unless explicitly stated.

Keywords: lossless compression; neural network weights; Forlais compression research; BF16; entropy ceiling; public compressor comparison; arithmetic coding; DEFLATE; LZMA; ZSTD; brotli

This is an approved public release of a Forlais Group academic paper.

Methodology Disclosure Boundary

This paper states the FCN evidence for the reported result at a public-safe abstraction level. It reports measured quantities, reported scope, and limitations. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

Lossless compression of trained neural network weights is studied across the FCN research programme as a coding question first and an engineering question second. For each FCN entry, the goal is to state the smallest public-safe claim that the evidence actually supports, with the format, scope, and reconstruction target made explicit.

This paper reports the FCN evidence for a wide comparison of public lossless compressors applied to BF16 weights of a publicly released large language model. The comparison is reported in aggregate; individual public compressor baselines (DEFLATE, LZMA, ZSTD) are reported in their own FCN entries and cited where appropriate. The motivation for the comparison is direct: if many public compressors converge to the same neighbourhood on this input, that is evidence the binding constraint lies in the source distribution and not in the back-end coder.

The headline numbers, sourced from the FCN evidence register, are a best reported ratio of approximately $1.5085\times$ on the tested setting from a tight single-tensor arithmetic coder, with the dispersion of the tested compressors falling within roughly half a multiplier of each other across more than sixty tested configurations. The companion FCN entry for the entropy ceiling reports approximately $1.522\times$ as the single-model bound, of which the best configuration captures approximately 99.1%.

The remainder of the paper states the public-safe scope, summarises the relevant background, presents the verification protocol and the cross-compressor results, summarises algebraic-randomness checks on the compressed bitstream, and records the limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- BF16 stored weights of a publicly released large language model;
- public lossless compressors at standard quality settings, with and without standard public pre-conditioners (bit-transpose, byte-swap, bit-plane separation, exponent grouping);
- single-model lossless compression with bit-exact reconstruction verified by SHA-256 round-trip;
- value-level entropy measurement of BF16 stored weights intended to detect residual structure;
- public-safe summarisation of measured quantities.

The result does not concern:

- lossy model compression, pruning, distillation, and quantisation-aware retraining as separate regimes;

- cross-model delta coding against a paired base model, which is reported in separate FCN entries and reaches different ratios;
- learned non-IID coders trained specifically on weight distributions;
- format redesign or architecture-specific compression that requires modifying model semantics;
- content-addressable deduplication, which is reported in separate FCN entries.

Each excluded regime is a distinct setting with its own evidence base. Ratios from those settings should not be compared with the headline number in this paper without first reconciling assumptions about coding scope, format, and reconstruction target.

3 Background and related work

Lossless compression of trained neural network weights sits at the intersection of classical source coding and contemporary neural network engineering. Shannon’s source coding theorem gives a lower bound on the expected code length of any lossless code in terms of the entropy of the source distribution [27, 4]. Huffman coding and arithmetic coding provide practical means to approach the entropy bound on a discrete source [14, 28, 26, 25]. Asymmetric numeral systems offer a related entropy-coding family with implementation properties between arithmetic and Huffman coding [7]. Universal compression of arbitrary sources via Lempel-Ziv dictionary substitution is the basis of most widely deployed general-purpose coders [29, 30].

The public compressors covered in this comparison are specified in their respective public references. DEFLATE, the back-end of gzip and zlib, is specified in IETF RFC 1951 [6]. LZMA is documented in the LZMA Software Development Kit [24]. ZSTD is standardised in IETF RFC 8878 [3]. Context-mixing methods including PPM and the ZPAQ family extend dictionary-based coders with multi-context predictors [2, 19]. Range coding underlies several of the back-ends in the comparison [20], as do Krichevsky-Trofimov-style universal codes within adaptive arithmetic coders [18].

Modern model weights are typically stored in BF16 or FP32. The relevant numeric formats are specified in the IEEE 754 standard and discussed in the BF16 and mixed-precision training literature [15, 17, 22]. We refer to the publicly released model artefacts associated with the architecture examined here [16].

Algebraic structure in finite-field representations is captured classically by linear-recurrence and shift-register synthesis [1, 21]. We use these tools to test whether a near-entropy compressed stream retains exploitable linear structure.

Adjacent lossy and learned-compression regimes for neural networks include pruning, quantised post-training inference paths, and distillation, all of which are explicitly outside the scope of this paper [10, 5, 9, 12, 8, 13]. An independent line of work has examined lossless coding of LLM weights using dynamic-length floating-point formats [11].

4 Verification protocol

The verification protocol that underlies the headline numbers is summarised below; lower-level implementation details remain internal.

- Input: BF16 tensors of a publicly released large language model, loaded directly from the published checkpoint format.

- **Preconditioners:** a fixed set of public preconditioners is applied independently before back-end coding, including identity, bit-transpose, byte-swap, bit-plane separation, exponent grouping, sign-magnitude rearrangement, and small zigzag and frequency-rank rearrangements. None of these preconditioners modifies the per-value entropy.
- **Back-end coders:** more than sixty public configurations spanning the DEFLATE, LZMA, ZSTD, brotli, range-coder, and ZPAQ-style families, at standard quality settings exposed by their public libraries.
- **Acceptance criterion:** a configuration is counted as successful only when the decoded byte stream reconstructs the original BF16 tensor exactly. Reconstruction is verified by SHA-256 round-trip against the original tensor bytes [23].
- **Reported ratios:** total raw bytes divided by total compressed bytes, aggregated across the tested tensors of the model and reported at the public-safe abstraction level.

The protocol is deliberately conservative. It uses only public components, accepts only bit-exact reconstruction, and reports a single aggregate ratio per configuration rather than per-tensor optima. The aim is to characterise the dispersion across the public-compressor landscape and to identify the best public configuration available on this input.

The set of preconditioners deserves explicit comment. None of them changes the per-value entropy of the input. They are storage-layout permutations whose role is to expose the underlying per-bit or per-component distributions to the back-end coder in a form the back-end can exploit. Some preconditioners help every back-end approximately equally (bit-transpose is the canonical example), while others help specific back-ends more than others (exponent grouping can interact badly with LZ77-style dictionaries when it produces unstable run lengths). The cross-compressor comparison therefore reports preconditioner-back-end pairs rather than back-ends in isolation; this is the same convention used in the public mega-tests that motivated the comparison.

5 Empirical results

The numerical summary in this section is reproduced from the internal FCN evidence register. Values appear here only at the public abstraction level. The release excludes underlying implementation details, run procedures, and source logs.

Reported quantity	Public-safe value
Number of public configurations tested	more than 60
Best aggregate ratio	approximately 1.5085x (tight arithmetic coder)
Best public-compressor configuration (LZMA family)	approximately 1.47x
Reference baseline (DEFLATE family)	approximately 1.437x
Dispersion across tested configurations	within roughly half a multiplier of each other
Single-model entropy ceiling (companion FCN entry)	approximately 1.522x
Best configuration relative to ceiling	approximately 99.1%
Reconstruction	bit-exact, SHA-256 verified
Numeric format	BF16

Table 1: Summary of public-safe FCN evidence for the cross-compressor comparison reported in this paper.

5.1 Family-level summary

Across the tested configurations, the cross-family ordering is consistent with the strength of each family's entropy coder. The tight arithmetic-coding configurations reach the best aggregate ratio at approximately $1.5085\times$, approaching the FCN single-model entropy ceiling. LZMA configurations cluster at approximately $1.46\text{--}1.47\times$. ZSTD at high levels reaches approximately $1.47\text{--}1.49\times$, and brotli at maximum quality lands in a similar neighbourhood. DEFLATE-based configurations cluster lower, at approximately $1.40\text{--}1.44\times$, consistent with Huffman coding's per-symbol quantisation overhead. A small number of configurations with poorly matched preconditioners produce ratios below $1.0\times$ on this input; these are documented in the internal log to characterise the failure modes of preconditioning, and they do not affect the upper-envelope conclusion.

5.2 Effect of preconditioning on the ordering

Holding the back-end constant and varying the preconditioner produces ordering effects that are themselves informative. On every strong back-end tested, the bit-transpose preconditioner raised the aggregate ratio by approximately $0.10\text{--}0.20$ over the raw byte stream, after which further preconditioners (bit-plane separation, exponent grouping with stable run lengths, sign-magnitude rearrangement) contributed at most a few additional percent. The diminishing returns are consistent with the bit-transpose recovering most of the back-end-visible structure in BF16 weights and the remaining preconditioners refining the distribution at the margin. Configurations with permutation-style preconditioners that introduced unstable run lengths produced lower aggregate ratios because the LZ77 dictionary stage could not amortise its matching cost over the resulting stream.

5.3 Algebraic-randomness check on the compressed bitstream

A separate algebraic-randomness check applies the Berlekamp-Massey linear-complexity test to a sample of the compressed bitstream produced by the best tested configuration [1, 21]. The measured linear complexity is consistent with the value expected from a maximum-entropy bit source of the same length, indicating that no short linear recurrence describes the compressed stream. We interpret this as evidence that the residual gap to the FCN entropy ceiling is implementation residue rather than exploitable linear structure left in the compressed output.

5.4 Interpretation

The clustering of more than sixty public configurations within a narrow band, with no configuration exceeding the FCN entropy ceiling, is the principal scientific content of this paper. It shows that the binding constraint in the single-model regime is the per-value entropy of the trained-weight distribution. Changing the back-end can move the configuration up or down within the band, but cannot move it past the ceiling. Improvements beyond the band require leaving the regime: cross-model conditioning, format change, lossy regimes, or specialised non-IID coders. We address each of these regimes in separate FCN entries.

6 Cross-architecture observation

The headline measurements in this paper are taken on a single publicly released large language model. The companion FCN entry that derives the entropy ceiling reports cross-architecture consistency of the ceiling itself: the per-value entropy of BF16 weights lies in a narrow band across the architectures tested in that entry, with the corresponding ceiling concentrated in the band $1.51\text{--}1.53\times$. The shape

of the cross-compressor envelope reported here is governed by the per-bit-position distribution rather than by architectural details, so the same envelope is expected to hold across the architectures inside the FCN band, with role-mix and exponent-spread differences absorbing small variation. We treat any cross-architecture measurement as warranting its own FCN entry rather than as a claim from the present paper.

7 Methodology context

The verification protocol used in this paper sits inside the broader FCN methodology for lossless compression evaluation. The protocol elements that matter for the present result are: (i) bit-exact reconstruction verified by SHA-256 round-trip on every reported configuration; (ii) reporting of aggregate ratios across the full tested tensor set rather than cherry-picked tensors; (iii) treating each preconditioner-back-end pair as a single configuration, so that the headline number is reproducible by stating both choices; and (iv) use of public-library implementations of all components named in this paper. The paper is intended to be cited and audited on its own terms; the internal operational log is decoupled from this public-safe report.

8 Connection to public literature

The clustering observed in the comparison is what classical source coding theory predicts for a near-IID source with fractional per-symbol entropy. The best practical coders on such sources differ by small amounts that reflect the tightness of their entropy back-end, not the cleverness of their dictionary stage. The measurements in this paper therefore corroborate the classical picture and add a specific quantitative anchor for the BF16 weight case: a band of approximately half a multiplier of dispersion, bounded above by the FCN entropy ceiling.

9 Implications

For practitioners storing BF16 model weights with bit-exact reconstruction, the practical implication is that the choice of public compressor matters only at the margin in the single-model regime. Reporting ratios above the ceiling for such regimes is a methodological red flag: in our experience these reports come from regime confusions, deduplication counting, or undisclosed lossy steps. We provide the comparison in this paper precisely so that future work has a defensible reference point for the public-compressor envelope on BF16 weights.

10 Discussion

The cross-compressor comparison closes a question that recurs in the field: whether a sufficiently clever choice of public compressor can exceed the entropy of the source on neural-network weights. The measurement shows that it cannot, within the single-model regime, for the tested architecture and numeric format. The dispersion observed across families is consistent with classical source coding theory and with the FCN entropy ceiling reported in a companion entry. We do not claim that the tested set is exhaustive; we claim only that the upper envelope is stable across more than sixty configurations and that no configuration exceeds the FCN entropy ceiling.

We also note three points that bear on how this comparison should be cited. First, the upper envelope is what should be cited, not the best run of any single configuration; cherry-picked per-tensor ratios

can sit slightly higher and are not informative about the general behaviour of the back-end. Second, the dispersion band reported here is specific to BF16 weights and to the role mix in the tested model; smaller models with heavier normalisation-layer mass or different attention-to-MLP ratios produce slightly different bands. Third, the comparison rules out only one direction of improvement: changing the public back-end. It does not rule out the orthogonal directions (cross-model conditioning, learned coders, format changes), which we evaluate separately.

11 Relation to other FCN entries

This paper is the cross-compressor anchor for the FCN single-model regime on BF16 weights of a publicly released large language model. The DEFLATE, LZMA, and ZSTD entries report individual baselines that sit inside the band reported here; the entropy-ceiling entry reports the upper bound of the band; and the closed-form Benford derivation provides the source-side explanation of why the band exists at the measured location. We refer readers interested in any single back-end to the corresponding FCN entry; the value of the present paper is the family-level dispersion picture and the upper-envelope conclusion.

12 Limitations

We note four limitations.

1. The reported ratios are empirical and apply to the specified architecture, numeric format, and tensor role mix; we do not claim universality beyond the tested networks.
2. The result is restricted to lossless, bit-exact reconstruction verified by SHA-256 round-trip. Lossy regimes are out of scope here.
3. The set of tested configurations is large but not exhaustive. Configurations not in the public-compressor landscape, including learned non-IID coders trained on weight distributions, are out of scope here.
4. Public reproducibility is limited by disclosure control; internal verification artefacts are maintained by Forlais Group but are not included in this paper.

13 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

Reference implementations and specifications for all public compressors named here are widely distributed; see the corresponding references for DEFLATE [6], LZMA [24], ZSTD [3], and the ZPAQ-family discussion [19].

14 Acknowledgements

We acknowledge internal review and pressure-testing of earlier internal review versions of the cross-compressor comparison, including correction of an earlier framing that quoted the best-of-sixty config-

uration without contextualising it against the FCN entropy ceiling. The reviewed version reports both the upper envelope and the ceiling so that the relationship between them is explicit.

15 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

16 Conclusion

The FCN evidence supports a verified, scoped conclusion: across more than sixty public lossless configurations spanning the DEFLATE, LZMA, ZSTD, brotli, and ZPAQ-style families applied to BF16 weights of a publicly released large language model, no tested configuration exceeds the FCN single-model entropy ceiling of approximately $1.522\times$. The best aggregate ratio is approximately $1.5085\times$, attained by a tight arithmetic coder and corresponding to approximately 99.1% of the ceiling. The dispersion across families is consistent with the strength of their entropy back-ends and with classical source coding theory. The result is reported here in its public-safe form and should be read alongside the limitations enumerated above.

We treat the upper-envelope conclusion as the principal scientific content of this paper. Future public coders that report single-model BF16 weight ratios materially above the FCN ceiling without changing the regime should be expected to specify their preconditioning, their bit-exact acceptance criterion, and the disposition of any side information; in our experience, ratios above the ceiling under proper accounting do not occur in this regime.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Expanded references, structured evidence summary, scope, and limitations.	Oliver Cooper
0.3	14 May 2026	Full evidence-grounded manuscript with public-safe abstraction, table summaries, and category-aware framing.	Oliver Cooper
1.0	14 May 2026	Publication-candidate manuscript: cross-family comparison framing, verification protocol, algebraic-randomness check, ceiling-relative interpretation, full limitations.	Oliver Cooper